Brightness Perceiving for Recursive Low-Light Image Enhancement

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***Abstract*—Due to the wide dynamic range in real low-light scenes, there will be large differences in the degree of contrast degradation and detail blurring of captured images, making it difficult for existing end-to-end methods to enhance low-light images to normal exposure. To address the above issue, we decompose low-light image enhancement (LLIE) into a recursive enhancement task and propose a brightness perceiving-based recursive enhancement framework for high dynamic range LLIE. Specifically, our recursive enhancement framework consists of two parallel subnetworks: adaptive contrast and texture enhancement network (ACT-Net) and brightness perception network (BP-Net). The ACT-Net is proposed to adaptively enhance image contrast and details under the guidance of the brightness adjustment branch and gradient adjustment branch, which are proposed to perceive the degradation degree of contrast and details in low-light images. To adaptively enhance images captured under different brightness levels, BP-Net is proposed to control the recursive enhancement times of ACT-Net by exploring the image brightness distribution properties. Finally, in order to coordinate ACT-Net and BP-Net, we design a novel unsupervised training strategy to facilitate the training procedure. To further validate the effectiveness of the proposed method, we construct a new dataset with a broader brightness distribution by mixing three low-light datasets. Compared with eleven existing representative methods, the proposed method achieves new state-of-the-art (SOTA) performance on six reference and no-reference metrics. Specifically, the proposed method improves the peak signal-to-noise ratio (PSNR) by 0.9 dB compared to the existing SOTA method.**

***Impact Statement*—The degrees of detail loss and contrast degradation vary in high dynamic low-light scenarios, making it challenging to reconstruct high-quality images. To address this issue, we propose a recursive enhancement framework that gradually improves contrast and detail texture while utilizing a brightness perception branch to control the extent of enhancement. Extensive experiments demonstrate that our proposed method outperforms existing methods and achieves remarkable generalization performance in real-world scenarios. Consequently, our method possesses prospective applications on intelligent displays and cameras to improve model robustness in complex and variable lighting conditions.**

Manuscript received 18 August 2023; revised 19 October 2023; accepted 19 November 2023. Date of publication 4 December 2023; date of current version 21 June 2024. This work was supported by the Natural Science Foundation of China No. 62206262, and National Key R&D Program of China under Grant 2020AAA0105700. This article was recommended for publication by Associate Editor Monjoy Saha upon evaluation of the reviewers' comments. *(Haodian Wang and Long Peng are the co-first authors.) (Corresponding authors: Yang Wang; Yang Cao.)*

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Digital Object Identifier 10.1109/TAI.2023.3339092

***Index Terms*—Detail enhancement, low-light image enhancement, recursive framework, unsupervised learning.**

I. Introduction

THE images captured under low illumination will suffer from statistical and structural properties distortion, resulting in low contrast, blur, and noise problems. This can significantly degrade the human visual effects and performance of high-level vision algorithms [3], [4], [5], [6], [7], [8]. Due to the high dynamic range of brightness in the real scene, the contrast degradation and detail texture distortion at different brightnesses are different, as shown in Fig. 1. Therefore, how to enhance the low-light images captured in wide dynamic range low-light scenes to normal exposures has received significant attention from researchers [1], [9].

In practice, one solution for this problem is to collect enormous paired low-light images from different light conditions and scenes and use these images to train an end-to-end enhancement network [2], [11]. However, directly using the end-to-end network often leads to under or over-exposure problems, as shown in the results of self-calibrated illumination (SCI) [2] in Fig. 2(c). This is because, due to the variations in statistical and structural exposure representations under different lighting conditions, the procedures for enhancing them to normal exposures differ greatly from each other. To generate better results, researchers propose to narrow the gap between different exposure representations. For example, researchers propose to map different exposure features to the exposure invariant feature space [12], [13]. However, learning the exposure invariant space through operations such as instance normalization will inevitably disrupt the structural features of images in different scenes. These methods are also limited by the representational capacity of the model, which makes it difficult to handle multiple low-light scenes, as shown in the results of fourier-based exposure correction (FEC) [1] in Fig. 2(c).

In this article, we decompose the low-light enhancement problem into a recursive enhancement task and propose a brightness perceiving recursive enhancement method for high dynamic range low-light image enhancement (LLIE). Specifically, a novel adaptive contrast and texture enhancement network (ACT-Net) is proposed as the enhancement network of the recursive framework (RF) to enhance the contrast and details of low-light images simultaneously. In ACT-Net, we introduce

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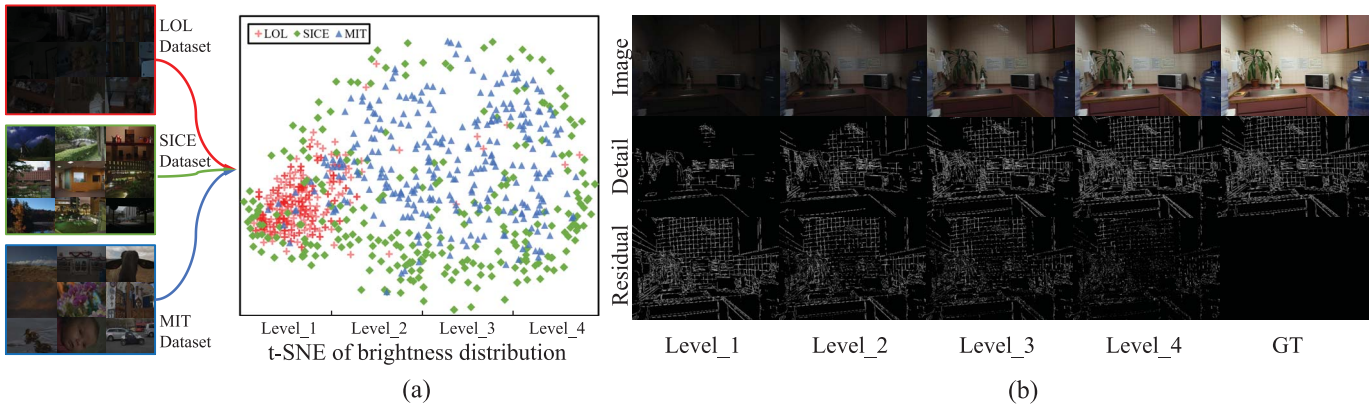

Fig. 1. (a) t-SNE [10] of brightness distribution of existing three prevailing low-light datasets. (b) The visualization of details degradation in different low-light scenes from Level_1 (darkest) to Level_4 (brightest). We can observe that the brightness distribution of low-light images varies significantly under different brightness levels; the lower the brightness, the greater the structural damage.

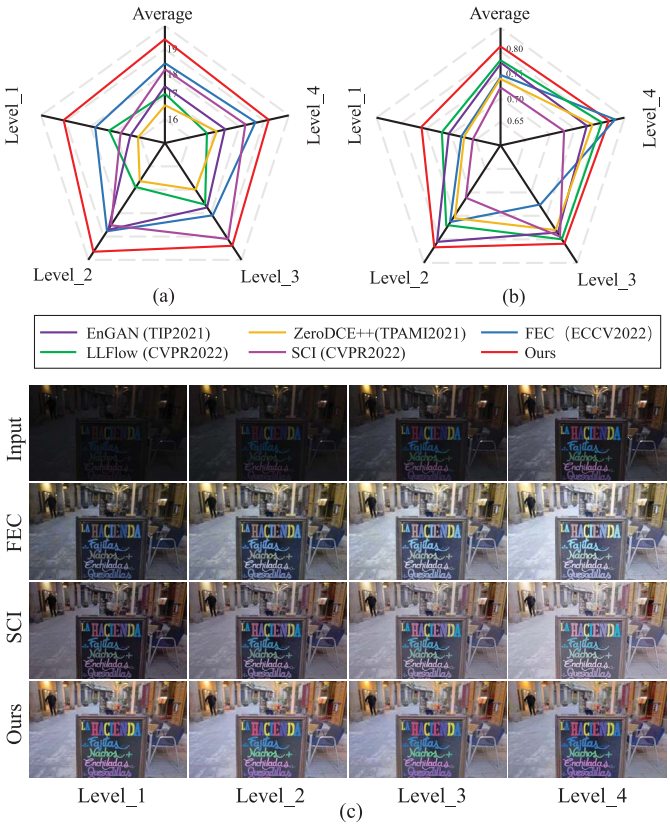

Fig. 2. (a) and (b) Comparison with existing methods on PSNR and SSIM under different brightness levels. (c) Visualization comparison with existing representative multiexposure enhancement method FEC [1] and low-light enhancement method SCI [2] in four different brightness of the same background.

central difference convolution (CDC) [14] convolutions to perceive high-frequency gradient information and introduce the brightness adjustment branch (BAB) and gradient adjustment branch (GAB) to balance image contrast and detail enhancement. To avoid under or over-exposure problems, we propose a novel brightness perception network (BP-Net) to predict the recursive factor of the enhancement network by perceiving the brightness distribution of the image and then utilize it to control the enhancement times of ACT-Net. Finally, to train the recursive enhancement framework, a novel unsupervised training strategy is proposed to obtain pseudolabels to train BP-Net, and joint fine-tuning ACT-Net and BP-Net to coordinate the contrast and detail enhancement of different areas of the image to obtain better visual effects. Experiments show that our method significantly outperforms the existing methods in PSNR and structual similarity index measurement (SSIM) metrics, as shown in Fig. 2(a) and (b).

The contributions can be summarized as follows:

1) Regarding the LLIE problem as an adaptive enhancement procedure, we propose a novel recursive enhancement framework for adaptive image enhancement by perceiving brightness distribution and balancing details and contrast enhancement.
2) A novel ACT-Net is proposed to enhance image contrast and structure simultaneously, which introduces CDC to balance details and brightness enhancement. Besides, to adaptively enhance images captured under different light levels, a novel BP-Net is proposed for perceiving brightness to control the recursive enhancement times of ACT-Net.
3) Comprehensive experiments demonstrate that, compared with the eleven LLIE and multiexposure correction methods, our proposed method achieves state-of-the-art performance in mixed low-light datasets with high dynamic ranges.

II. Related Work

A. Low-Light Image Enhancement

Low-light image enhancement aims to automatically improve the visibility of images captured under low-light conditions, which can be divided into traditional methods and learning-based methods [15], [16], [17], [18], [19]. Most traditional methods usually utilize histogram equalization [20], [21], [22] and Retinex theory [23], [24], [25], [26] to enhance image contrast and details. However, since these methods heavily rely on handcrafted image priors, they usually cannot generate satisfactory results in real-world images [15], [27].

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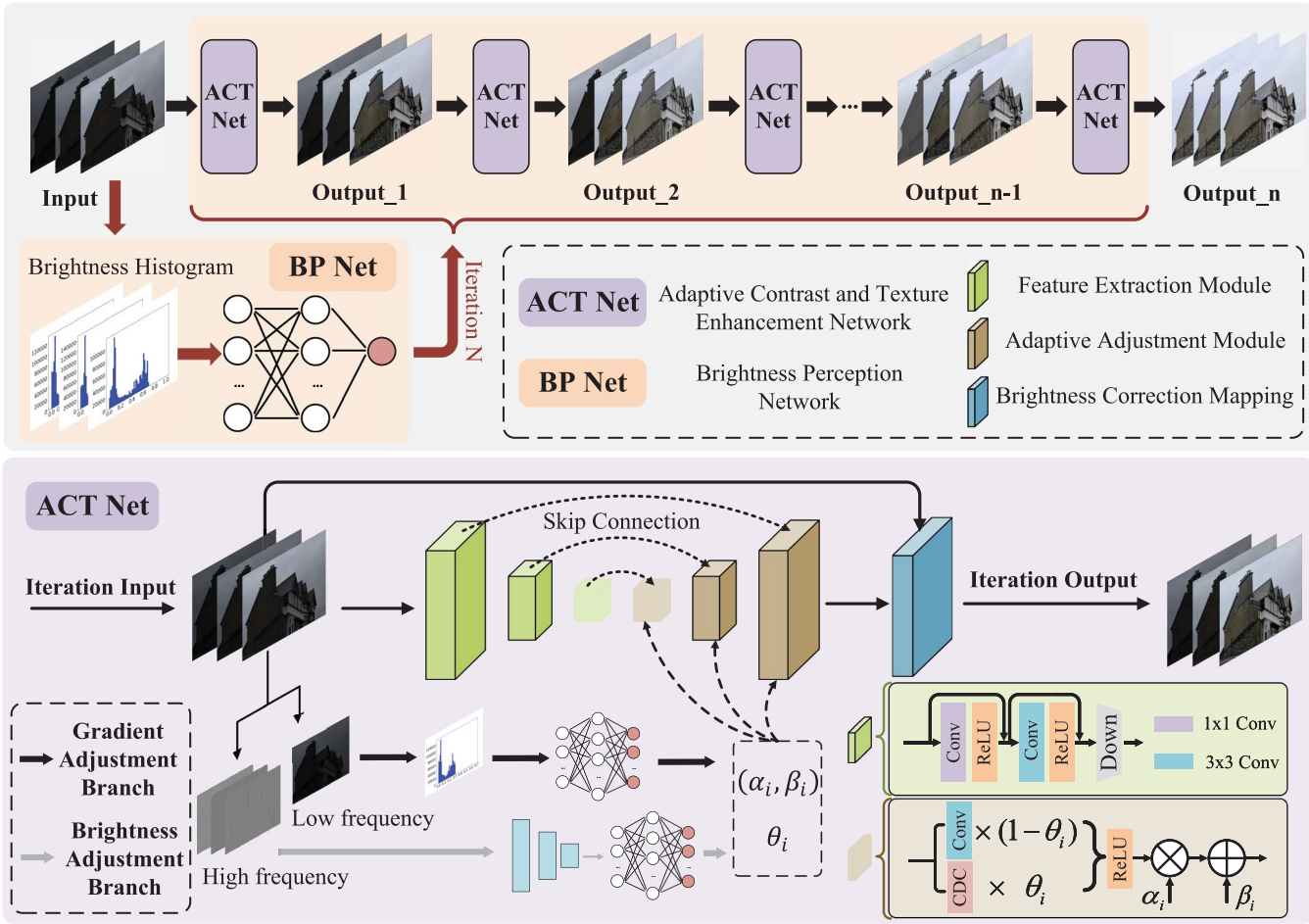

Fig. 3. Overall architecture of our recursive enhancement framework, which consists of the ACT-Net and BP-Net.

In recent years, with the development of deep learning, many learning-based LLIE methods have been proposed, which have made significant progress [11], [28], [29], [30], [31], [32]. Wei et al. [27] propose RetinexNet to estimate the illumination map and enhance the low-light images. Wang et al. [33] propose an end-to-end image-to-illumination mapping network to enhance under-exposed photos. Wu et al. [11] propose a Retinex-based deep unfolding network that unfolds the LLIE problem into a learnable network to decompose a low-light image into reflectance and illumination layers. However, most of the above learning-based LLIE methods rely on paired data for supervised training, which makes it hard to collect a large-scale dataset in practical environments. Therefore, unsupervised LLIE methods have attracted a lot of attention. For example, Jiang et al. [34] propose an unsupervised EnGAN with a global-local discriminator and UNet structure generator for LLIE. Ma et al. [2] propose an unsupervised self-calibrated illumination learning framework for efficiently enhancing low-light images. However, most existing LLIE methods can only enhance low-light images in specific low-light conditions and are difficult to adapt to real scenes with high dynamic low-light ranges.

B. Multiexposure Correction

Multiexposure correction (MEC) aims to learn a unified network for enhancing images captured under and over-exposure conditions [1], [12], [35], [36]. Afifi et al. [35] propose a pyramid structure network to enhance under and over-exposure images to normal exposure in a coarse-to-fine manner. Huang et al. [1] propose a deep Fourier-based exposure correction network to reconstruct the representation of image lightness and structure components progressively. Although these methods can enhance the high dynamic range image caused by improper exposure to a certain level of brightness, however, due to the lack of brightness perception of the original image, they cannot further accurately control the brightness of the enhanced image to produce pleasant results.

III. Recursive Low-Light Enhancement Network

In this section, we introduce the proposed brightness perceiving recursive enhancement framework, which consists of the ACT-Net and BP-Net, as shown in Fig. 3. The BP-Net perceives the brightness distribution of the input low-light images and predicts the recursive factor for jumping out of the recursive process. Meanwhile, we design a novel unsupervised training strategy to facilitate the training procedure. More details are described below.

A. Brightness Perception Network

In order to adaptively enhance image contrast and avoid under or over-exposure problems, we propose the BP-Net to

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estimate the recursive enhancement times of ACT-Net, as shown in Fig. 3. First, in order to remove the influence of color information and reduce redundant information for brightness perception, we translate the input low-light image from RGB color space to HSV space and only reserve the brightness information in the V channel. Secondly, we calculate the histogram of this V channel to obtain brightness statistics information. Finally, BP-Net utilizes the statistics histogram as input to estimate recursive enhancement times and is composed of three FC-ReLU layers and a Sigmoid activate function. The BP-Net is trained by the following loss function:

$$L_p = \left|\mathrm{BP}(x) - \mathrm{Label}_p\right|. \tag{1}$$

Where x represents the input low-light image, Label_p represents the pseudo label obtained by the unsupervised training strategy in Section III-D. The max and min output iterations of BP-Net are defined as $\rho_{\max}$ and $\rho_{\min}$, which means $\mathrm{BP}(x) \in [\rho_{\min}, \rho_{\max}]$.

B. Adaptive Contrast and Texture Enhancement Network

We propose a novel ACT-Net to recursively enhance the details and contrast of low-light images captured in high dynamic range environments. ACT-Net uses the UNet-style structure network as the baseline, including feature extraction modules, adaptive adjustment modules (AAMs), and brightness correction modules, as shown in Fig. 3. For feature extraction modules, we use 1×1 and 3×3 convolutional networks with ReLU to extract image features and add a downsampling layer to reduce the spatial scale. To effectively enhance the contrast and details of the image during the feature decoding, we designed the AAM to embed brightness and gradient information into the decoder, thereby performing feature modulation on the middle layer.

Specifically, we generate affine transformation parameters by extracting image brightness as prior information to modulate the middle layer features of the network and adaptively correct brightness. Additionally, we designed a BAB (four fully connected layer and rectified linear unit (FC-ReLU) layers) to extract the corresponding brightness modulation factors α and β. In order to enhance the image's texture structure while correcting brightness, we introduced central differential convolution (CDC) [14] to embed gradient information, which has been proven to extract robust information independent of lighting. We also designed a GAB to obtain the gradient adjustment coefficient θ to automatically balance brightness and texture enhancement, which consists of three cascaded 3×3 convolutions with ReLU.

To avoid the interference of low-frequency components, we decompose the input of each iteration into high- and low-frequency components through haar wavelet decomposition and employ the high-frequency components instead of the original image as the input of the GAB. Meanwhile, we extract a brightness statistical histogram from the low-frequency component of the image as input to the BAB, which means converting the input from the spatial domain into the statistical domain to avoid further interference of redundant information. AAM can be expressed as the following:

$$\begin{aligned}\mathrm{AAM}_i(f) = \mathrm{ReLU}(&\theta_i \cdot \mathrm{CDC}(f) \\ &+ (1-\theta_i) \cdot \mathrm{Conv}(f)) \cdot \alpha_i + \beta_i\end{aligned} \tag{2}$$

where AAM_i represents the ith the AAMs, f represents the input feature, θ_i represents the gradient adjustment coefficient, α_i and β_i represent brightness adjustment factors.

Finally, we map the output of the ACT-Net to image space by brightness correction mapping [9], which can be expressed as follows:

$$E_n(x) = E_{n-1}(x) + \mathrm{ACT}(x) \cdot E_{n-1}(x) \cdot (1 - E_{n-1}(x)) \tag{3}$$

where x represents input low-light images, n is the number of iterations, E_n represents the enhanced result of the nth iteration. E_0 represents the original input low-light image.

In order to effectively improve the contrast, suppress the color deviation, and maintain the smoothness of the image during enhancement, we use three representative loss functions for training ACT-Net in an unsupervised manner:

To control the exposure level of the enhanced image, we use exposure control loss $L_{\exp}$

$$L_{\exp} = \frac{1}{R} \cdot \sum_{k=1}^{R} \left\|\bar{V}_k(\mathrm{ACT}(x)) - E\right\| \tag{4}$$

where $\mathrm{ACT}(x)$ represents the image enhanced by ACT-Net, V is the nonoverlapping local patch, patch size of V is 16×16, $\bar{V}_k$ represents the average brightness value of the kth local patch, R represents the number of V_k, E is manual exposure from [9], which is set to 0.6 [1], and $\mathrm{ACT}(x) \in [0, 1]$.

To suppress the potential color deviations in the enhanced image, we use color constancy loss L_{col}

$$L_{\mathrm{col}} = \sum_{\forall(p,q)\in\varepsilon} \left(J^p - J^q\right)^2, \varepsilon = \{(R,G),(R,B),(G,B)\} \tag{5}$$

where J^p denotes the average intensity value of p channel in the enhanced image, (p, q) represents a pair of channels. R, G, and B represent the red, green, and blue channels in the RGB color space, respectively.

We use illumination smoothness loss L_{tv_m} to preserve the monotonicity relations between neighboring pixels in the output of ACT-Net:

$$L_{\mathrm{tv}_m} = \frac{1}{N} \sum_{n=1}^{N} \sum_{c\in\xi} \left(|\nabla_x m_n^c| + \nabla_y m_n^c\,|\right)^2, \xi = \{R, G, B\}. \tag{6}$$

where m represents output mapping feature of ACT-Net, N is the number of iteration, ∇_x and ∇_y represent the horizontal and vertical gradient operations, respectively.

[1] We have carried out statistical experiments on many natural images under normal exposure, and observed that most of the average brightness of these images is close to 0.6.

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C. Total Loss Function

We use the following L_{total} for training:

$$L_{\text{total}} = \lambda_{\text{exp}} \cdot L_{\text{exp}} + \lambda_{\text{col}} \cdot L_{\text{col}} + \lambda_{tv_m} \cdot L_{tv_m} + \lambda_p \cdot L_p. \quad (7)$$

where λ_{exp}, λ_{col}, λ_{tv_m}, and λ_p are tradeoff factors. In this article, we set λ_{exp} to 1, λ_{col} to 0.5, λ_{tv} to 200, and λ_p to 0.001, respectively.

D. Proposed Unsupervised Training Strategy

To effectively train the proposed framework, we propose a novel unsupervised training strategy consisting of the following three steps:

1) *Pretrain ACT-Net*. We only use low-light images with brightness Level_4 to train the ACT-Net, manually set the recursive enhancement times of the ACT-Net to 1, and train the ACT-Net in an unsupervised manner through the loss functions L_{exp}, L_{col}, and L_{tv_m} in Section III-B. After training, the ACT-Net has the ability to enhance Level_4 low-light images to normal exposure images.
2) *Pretrain BP-Net*. We freeze the parameters of the ACT-Net to obtain Label_p, which is used to train the BP-Net. Specifically, for images from Level_4, we set Label_p to 1; For images from Level_1 to Level_3, we use ACT-Net to recursively enhance images until the average brightness of the enhanced image is approximately equal to 0.6, and then we record this recursive enhancement times as Label_p for training BP-Net. We optimize the parameters of BP-Net by the L1 loss between BP-Net output and Label_p, as described in Section III-A.
3) *Fine-tuning overall network*. Since the manual normal exposure of 0.6 [9] is a hard global threshold, it is difficult to enhance the brightness of different local image patches adaptively. Therefore, we utilize images of all brightness levels to fine-tune the overall network by L_{total}.

To make it easier to understand our proposed unsupervised training strategy in the manuscripts, we present a detailed pseudocode of the proposed unsupervised training strategy, as shown in Algorithm 1.

Algorithm 1 Unsupervised Training Strategy

Require: low light image divided into $\{Level_1, Level_2, Level_3, Level_4\}$
Step1: pre-train ACT-Net
Input: $\{Level_4\}$
for image in $\{Level_4\}$ **do**
  optimize ACT-Net by unsupervised image enhancement loss(ACT-Net(image))
end for
Step2: pre-train BP-Net
Input: $\{Level_1, Level_2, Level_3, Level_4\}$
freeze the parameters of ACT-Net
for image in $\{Level_1, Level_2, Level_3, Level_4\}$ **do**
  iter_num = 0
  while global brightness average of image < 0.6 **do**
    image = ACT-Net(image)
    iter_num += 1
  end while
  pseudo-label = iter_num
  train BP-Net by calculate L1Loss(BP-Net(image),pseudo-label)
end for
Step3: fine-tuning overall network
Input: $\{Level_1, Level_2, Level_3, Level_4\}$
for image in $\{Level_1, Level_2, Level_3, Level_4\}$ **do**
  jointly optimizes ACT-Net and BP-Net by total loss
end for

IV. Experiments and Analysis

A. Experiment Setting

1) Datasets: We mix three prevailing low-light enhancement datasets for training and evaluation.

Specifically, we mix low-light images from LOL [27], SICE [37], and MIT-Adobe FiveK [38] datasets, calculate the average brightness of each image on the V channel in HSV color space and then divide these images according to the images' average brightness range from $[0, 0.15)$, $[0.15, 0.3)$, $[0.3, 0.45)$ to $[0.45, 0.6]$ (max average brightness is lower than 0.6), denoted as Level_1 to Level_4, respectively. Finally, we perform a large low-light range dataset with high dynamic range, including 1167 images of Level_1, 1382 images of Level_2, 1130 images of Level_3, and 889 images of Level_4, and the remaining 84 images for testing.

To explore the generalization of the proposed method in real scenarios, we conduct further testing on five unpaired low-light image datasets, including NPE [39], low-light image enhancement dataset (LIME) [40], multi-exposure image fusion dataset (MEF) [41], digital image contrast enhancement commercial dataset (DICM) [42], and vasileios vonikakis dataset (VV)[2].

2) Training Setting: In our experiment, we use the Adam optimizer with a batch size of 8 and an image patch size of 512×512 to train our methods on one NVIDIA 1080Ti GPU. The learning rate is set to 0.0001, the total epoch is set to 200, and the $\rho_{\max}$ and $\rho_{\min}$ are set to 10 and 1.

3) Comparison Methods: We compare the proposed method with five state-of-the-art unsupervised LLIE methods (including SCI [2], SSIENet [43], ZeroDCE++ [9], RUAS [44], and EnGAN [34]), four advanced supervised learning LLIE methods (including RetinexNet [27], KinD [45], DRBN [46], and LLFlow [47]) and two MEC methods (including multi-scale photo exposure correction (MSEC) [35] and FEC [1]).

4) Metrics: To comprehensively demonstrate the superiority of our method, we use three reference metrics and three no-reference metrics to evaluate the performance. For reference metrics, we use the PSNR ↑[3] [48], SSIM ↑ [49] and LPIPS ↓[2] [50]. For no-reference metrics, we use NIQE ↓ [51], EME ↑ [52], and LOE ↓ [39] for evaluation.

[2] https://sites.google.com/site/vonikakis/datasets
[3] ↑ means the higher, the better, ↓ means the lower, the better.

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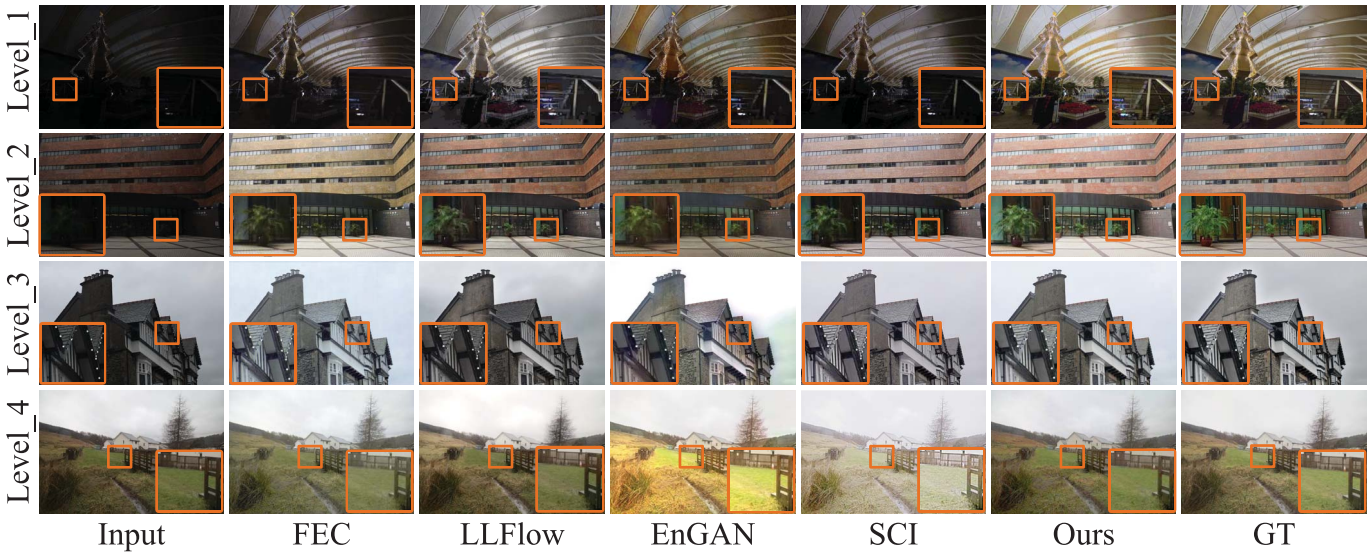

Fig. 4. Qualitative comparison with existing state-of-the-art methods. From the top to bottom row are the low-light conditions from Level_1 to Level_4, where Level_1 represents the darkest low-light images, and Level_4 represents the brightest.

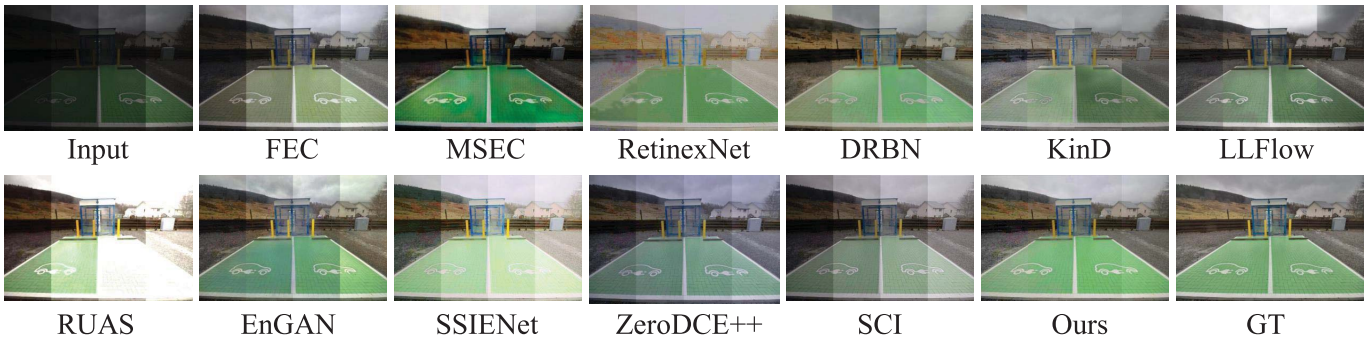

Fig. 5. Comparison of enhancement results in the same scene with existing representative methods. From the left to right in each image are the low-light conditions from darkest to brightest.

B. Qualitative Results

Fig. 4 shows the qualitative comparison with existing SOTA methods in different low-light conditions from Level_1 to Level_4. We can observe that in low-light scenes with high dynamic ranges, images enhanced by our proposed method are superior to the existing methods in image detail and contrast. For example, in the darkest scene, as shown in the top row in Fig. 4, the existing SOTA methods are unable to effectively enhance the contrast of the image, and the details of the enhanced image are lost, while our method can enhance the low-light image to the same brightness distribution as ground truth (GT), and also enhance the local texture details of the image, as shown in the red box. This demonstrates the effectiveness of our proposed ACT-Net. Besides, we can observe that in the brightest scene Level_4, images enhanced by EnGAN [34] and LLFlow [47] appear color distortion, and images enhanced by SCI [2] appears over-exposure problem causing the image to be overexposed, as shown in the bottom row in Fig. 4. However, our proposed method can utilize BP-Net to perceive image brightness and then assist ACT-Net to adaptively enhance the low-light image to the normal brightness range, which improves image details and contrast at the same time without color distortion and over-exposure. It demonstrates that the proposed method can effectively adapt to natural scenes with varying brightness levels without additional fine-tuning.

To intuitively show the effectiveness of the proposed method, Fig. 5 showcases the enhancement results of various methods on the same scene under different lighting conditions. It can be observed that most methods exhibit under or over-exposure when dealing with different lighting scenarios. In comparison, the proposed method enhances the image more uniformly and closely to the GT.

To further demonstrate that the proposed method can adaptively enhance image contrast, as shown in Fig. 6(a), we perform the brightness histogram (BH) t-SNE visualization of the enhanced image. We can observe that the images enhanced by SCI and FEC are still under-exposure/over-exposure, causing brightness distribution to be discrete. However, the brightness distribution of images enhanced by our method is clustered and closer to GT. To demonstrate the superiority of the proposed ACT-Net in image detail enhancement, we present the image

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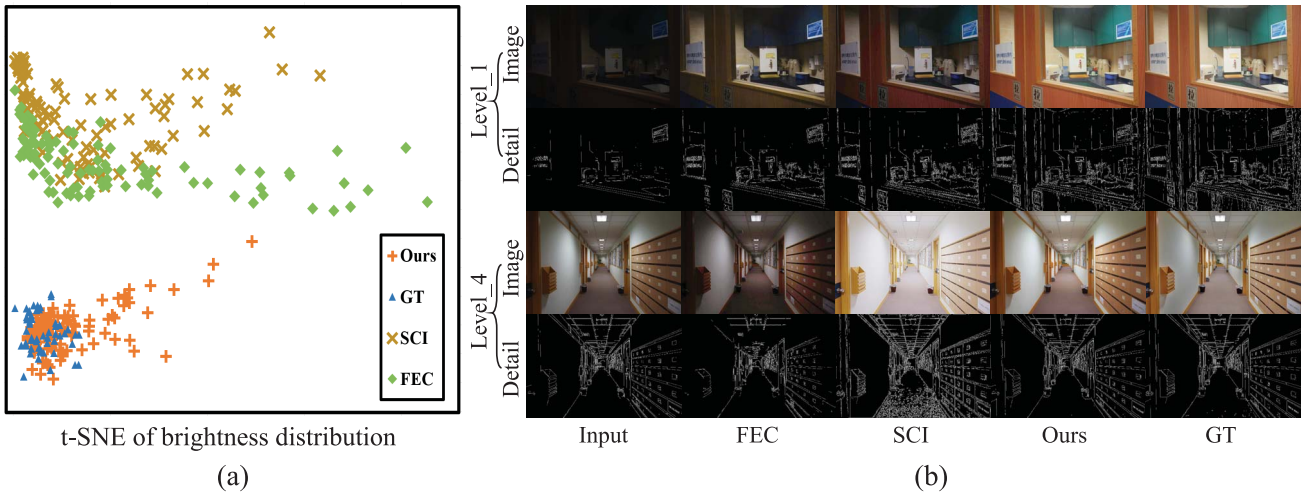

Fig. 6. (a) t-SNE visualization of brightness distribution of enhanced results. (b) The comparison of image detail enhancement result. The details are extracted by the Canny edge detector.

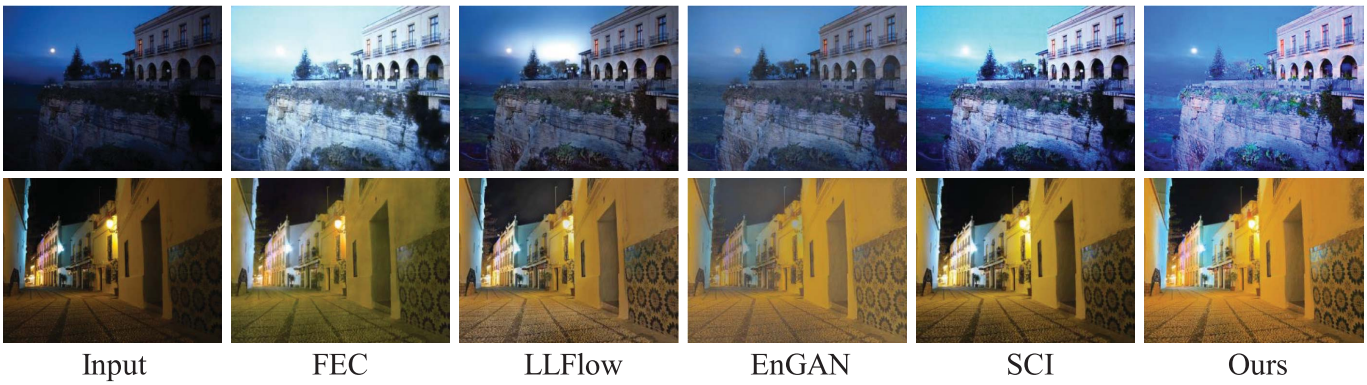

Fig. 7. Comparison with existing representative methods on five representative real-world datasets (including NPE, LIME, MEF, DICM, and VV).

detail enhancement results in low-light Level_1 and Level_4 scenes, as shown in Fig. 6(b). It can be easily observed that the detail texture enhanced by our proposed method is closer to GT.

Furthermore, to evaluate the generalization ability of the proposed method in real night scenes, we directly evaluate our method on five real datasets (including NPE [39], LIME [40], MEF [41], DICM [42], and VV). Note that all comparison methods are retrained on the same datasets as ours and directly tested on five other datasets without fine-tuning. As shown in Fig. 7, images enhanced by FEC [1] appear over-exposure caused by over-enhancement, and the images enhanced by LLFlow [47] exhibit unnatural light spots in some scenes. In contrast, our proposed method enhances low-light images more visually pleasuring.

Nevertheless, our method also has some limitations in dealing with the noise in real low-light images. As shown in Fig. 8, our enhanced results have a noise residual due to the absence of a denoising module in our proposed method. And it is a crucial direction for improvement in our future work.

Fig. 8. Examples of failed results with noise residue.

C. Quantitative Results

In Table I, our proposed method is compared with eleven SOTA methods on PSNR/SSIM metrics at different brightness levels. One can observe that our proposed method achieves the best performance in the majority of low-light environments.

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TABLE I
PSNR ↑ AND SSIM ↑ RESULTS FOR ELEVEN COMPARISON METHODS ACROSS FOUR DIFFERENT BRIGHTNESS LEVELS ON OUR CONSTRUCTED DATASETS (INCLUDING LOL [27], SICE [37], AND MIT-ADOBE FIVEK [38]). THE BEST AND SECOND-BEST PERFORMANCES ARE MARKED AS **BOLD** AND **BOLD**, RESPECTIVELY

Methods	Level_1		Level_2		Level_3		Level_4		Average	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
MSEC	17.056	0.671	18.736	0.689	**19.291**	0.771	18.826	0.724	**18.477**	0.714
FEC	17.791	0.685	18.938	0.767	18.166	0.738	**18.867**	**0.842**	18.440	0.758
RetinexNet	**18.362**	0.731	18.524	0.776	16.813	0.766	14.865	0.760	17.141	0.758
DRBN	16.161	0.658	16.385	0.683	16.592	0.713	16.851	0.734	16.498	0.697
KinD	17.276	0.715	17.881	0.790	18.726	0.795	17.292	0.756	17.794	0.764
LLFlow	17.482	0.723	16.969	0.770	17.666	**0.803**	16.867	0.806	17.246	**0.775**
EnGAN	16.488	0.704	18.907	**0.812**	17.891	0.793	17.482	0.772	17.692	0.771
RUAS	13.919	0.661	14.581	0.676	10.974	0.642	9.790	0.543	12.316	0.630
SSIENet	17.042	**0.781**	15.989	0.806	13.898	0.755	10.969	0.728	14.475	0.768
ZeroDCE++	16.099	0.683	16.721	0.759	17.035	0.791	17.121	0.794	16.744	0.757
SCI	16.883	0.655	**18.787**	0.710	19.133	0.796	18.381	0.734	18.296	0.724
Ours	**19.065**	**0.752**	**19.702**	**0.819**	**19.435**	**0.813**	**19.193**	**0.822**	**19.349**	**0.802**

TABLE II
COMPARISON WITH NINE EXISTING LOW LIGHT ENHANCEMENT METHODS ON OUR CONSTRUCTED DATASETS BY THREE NO-REFERENCE AND THREE REFERENCE METRICS. THE BEST AND SECOND-BEST PERFORMANCES ARE MARKED AS **BOLD** AND **BOLD**, RESPECTIVELY

Metrics	Supervised Method				Unsupervised Method					
	RetinexNet	DRBN	KinD	LLFlow	EnGAN	RUAS	SSIENet	ZeroDCE++	SCI	Ours
PSNR↑	17.14	16.50	17.79	17.25	17.69	12.32	14.47	16.74	**18.30**	**19.34**
SSIM↑	0.76	0.70	0.76	**0.78**	0.77	0.63	0.77	0.76	0.72	**0.80**
LPIPS↓	0.20	0.24	0.18	**0.16**	0.17	0.22	0.18	0.16	0.18	**0.14**
LOE↓	745.99	555.75	448.32	458.45	566.22	576.19	422.22	448.11	**408.81**	**398.94**
NIQE↓	2.65	2.92	2.53	**2.41**	2.77	4.61	3.05	2.41	2.49	**2.39**
EME↑	4.17	5.17	5.09	5.49	**5.86**	4.73	4.59	5.13	5.25	**5.68**

TABLE III
COMPARISON OF THE AVERAGE PERFORMANCE OF EXISTING ELEVEN METHODS ON FIVE REPRESENTATIVE REAL-WORLD DATASETS (INCLUDING NPE, LIME, MEF, DICM, AND VV) BY THREE NO-REFERENCE METRICS. THE BEST PERFORMANCE AND SECOND BEST PERFORMANCE ARE MARKED AS **BOLD** AND **BOLD**, RESPECTIVELY

Metrics	Supervised Method						Unsupervised Method					
	FEC	MSEC	RetinexNet	DRBN	KinD	LLFlow	EnGAN	RUAS	SSIENet	ZeroDCE++	SCI	Ours
LOE↓	479.57	581.05	793.68	673.60	464.32	431.48	549.43	541.71	407.96	385.37	**376.92**	**363.58**
NIQE↓	3.52	3.75	3.91	3.92	3.72	3.69	**3.42**	4.96	4.40	3.53	3.84	**3.40**
EME↑	5.77	6.18	3.78	5.99	6.57	6.73	**6.81**	5.68	5.12	6.49	6.63	**6.77**

For example, compared with the existing supervised and unsupervised methods, our method outperforms the PSNR and SSIM performances by 0.915 and 0.007 under the Level_2 low-light environment. This is because our adaptive RF can perceive image brightness and perform accurate enhancement without under or over-exposure. Besides, compared with the existing unsupervised SOTA method SCI [2], our proposed unsupervised method outperforms SCI by a large margin under all low-light environments. This demonstrates that the proposed method can be better applied to the high dynamic range low-light environment in real scenes.

To comprehensively compare the quality of enhanced images, we perform image quality comparisons on three reference metrics (PSNR↑, SSIM↑, and LPIPS↓) and three no-reference metrics (LOE↓, NIQE↓, and EME↑) in all brightness environments, as shown in Tables II and IV. One can observe that for the average performance in all brightness environments, we achieve SOTA performance on the five image quality metrics, including PSNR, SSIM, LPIPS, LOE, and natural image quality evaluator (NIQE), and outperform existing LLIE methods by 1.04 on PSNR, 0.02 on SSIM, 0.02 on LPIPS, 9.87 on LOE, and 0.02 on NIQE. We also outperform two SOTA multiexposure correction methods, as shown in Table IV. This demonstrates that the images enhanced by our proposed method are more natural and high-quality.

TABLE IV
COMPARISON WITH TWO MULTIEXPOSURE CORRECTION METHODS ON OUR CONSTRUCTED DATASETS BY THREE NO-REFERENCE AND THREE REFERENCE METRICS. THE BEST PERFORMANCE IS MARKED AS **BOLD**

	PSNR↑	SSIM↑	LPIPS↓	LOE↓	NIQE↓	EME↑
MSEC	18.48	0.71	0.20	525.72	2.48	5.64
FEC	18.44	0.76	0.18	618.79	2.68	4.56
Ours	**19.35**	**0.80**	**0.15**	**398.94**	**2.39**	**5.68**

As shown in Table III, we can observe that our method outperforms existing methods in most cases on real-world

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TABLE V
COMPARISON WITH EXISTING METHODS ON COMPUTATIONAL COST. THE BEST PERFORMANCE AND SECOND BEST PERFORMANCE ARE MARKED AS **BOLD** AND **BOLD**, RESPECTIVELY

Methods	FLOPs(G)	Params(M)	Time(S)	PSNR↑	SSIM [49]↑
SCI [2]	0.0620	0.000348	0.001688	18.30	0.72
ZeroDCE++ [9]	5.2112	0.078912	0.002375	16.74	0.76
RUAS [44]	0.2813	0.001437	0.042659	12.32	0.63
DRBN [46]	37.7902	0.577168	0.065768	16.50	0.70
KinD [45]	29.1303	8.540217	0.135406	17.79	0.76
EnGAN [34]	61.0102	8.636037	0.135406	17.69	0.77
SSIE [43]	34.6070	0.682400	0.150337	14.47	0.77
RetinexNet [27]	136.0151	0.838364	0.346362	17.14	0.76
FEC [1]	5.8156	0.151910	0.601818	**18.44**	0.76
LLFlow [47]	26.3169	38.86652	0.717524	17.25	**0.78**
Ours	3.6633	1.140580	0.179724	**19.35**	**0.80**

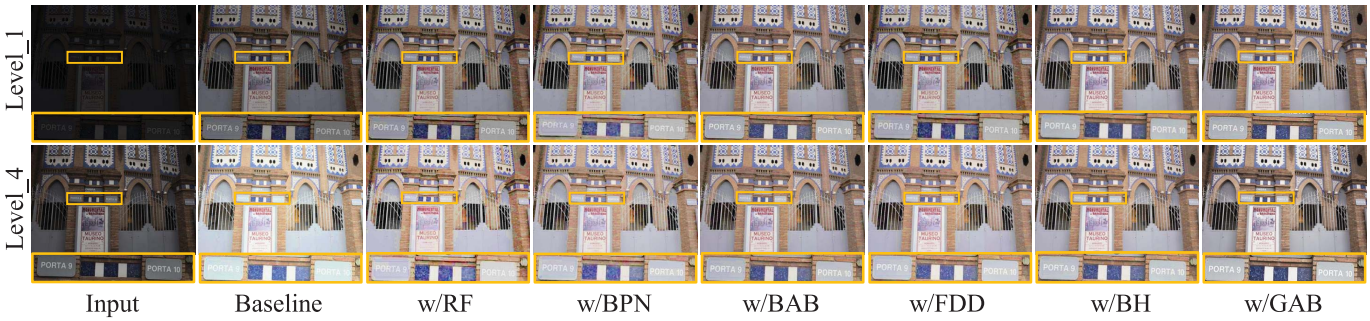

Fig. 9. Visual comparison of ablation studies. From left to right are the baseline and results after add RF, BPN, BAB, FDD, BH, and GAB sequentially on the basis of the baseline model.

TABLE VI
ABLUTION STUDY RESULTS. √ AND × REPRESENT WITH AND WITHOUT, RESPECTIVELY

	Ablation Study Setting						Performance	
	w/RF	w/BPN	w/BAB	w/FDD	w/BH	w/GAB	PSNR↑	NIQE↓
(a)	×	×	×	×	×	×	17.285	4.253
	√	×	×	×	×	×	18.441	3.882
	√	√	×	×	×	×	18.771	2.613
(b)	√	√	√	×	×	×	18.916	2.578
	√	√	√	√	×	×	19.091	2.527
	√	√	√	√	√	×	19.243	2.488
	√	√	√	√	√	√	**19.349**	**2.389**

datasets. Specifically, we achieve state-of-the-art performance on two no-reference image quality metrics, outperforming the existing LLIE methods by 13.34 on LOE and 0.02 on NIQE, respectively.

D. Computational Cost Comparison

In order to better demonstrate the superiority of the proposed method, we compare the computational cost with the existing method, and the results are shown in Table V. We can observe that our proposed method not only achieves state-of-the-art performance but also has a great advantage in the computational cost, compared with some large LLIE models such as LLFlow [47] and FEC [1].

E. Ablation Study

To demonstrate the effectiveness of the proposed method, we conduct comprehensive ablation studies:

1) Proposed framework: RF and BPN: We conduct ablation studies on RF and BPN, as shown in Table VI(a). Note that the first row in Table VI means baseline (i.e., UNet). After using RF, the input low-light images are enhanced by baseline until the average brightness is above 0.6, which improves the performance of the baseline by 1.16 and 0.371 on the PSNR and NIQE. After using BPN, the baseline can adaptively enhance input low-light images, avoiding under or over-exposure problems, which further improves the performance of the baseline by 0.33 and 1.269 on the PSNR and NIQE. As shown in Fig. 9, compared with the baseline, w/RF and w/BPN can effectively avoid the over-exposure problem, demonstrating that proposed RF and BPN can improve the quality of the enhanced image.

To verify the generality of the proposed framework, we replaced the backbone of our method with ZeroDCE++ [9], which is also a classical unsupervised training backbone, denoted as DCE++(ours). As shown in Fig. 10, we can observe that compared with DCE++, the contrast enhancement and detail enhancement capabilities of DCE++(ours) have been significantly improved in different brightness scenes. It demonstrates that our proposed framework can be effectively generalized to other low-light enhancement backbones and significantly improve their performance.

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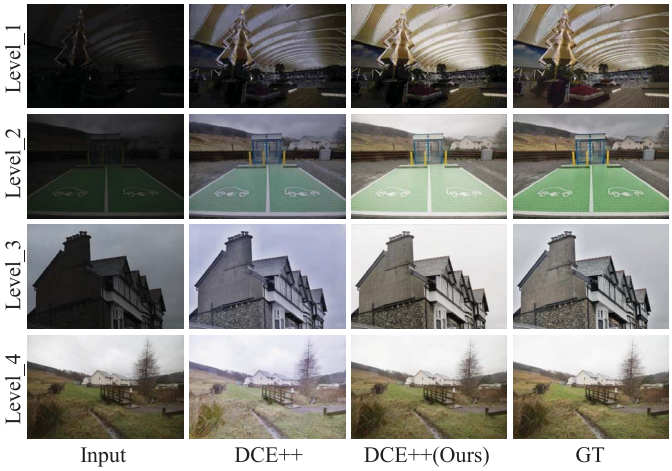

Fig. 10. Visual comparison on DCE++ and DCE++ (ours).

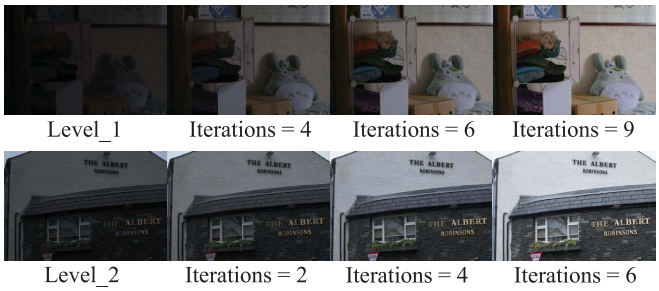

Fig. 11. Visualization of intermediate enhancement results under Level_1 and Level_2 low-light environments.

TABLE VII
ABLUTION STUDY RESULTS FOR THE LOSS FUNCTION, THE BEST PERFORMANCE IS MARKED AS **BOLD**

w/	PSNR↑	SSIM↑	LPIP↓	LOE↓	NIQE↓	EME↑
$L_p + L_{\text{tv}}$	12.67	0.66	0.31	613.72	2.82	4.71
$L_p + L_{\text{exp}}$	17.32	0.71	0.23	525.13	2.65	4.09
$L_p + L_{\text{exp}} + L_{\text{color}}$	18.19	0.75	0.21	422.37	2.57	5.43
$L_p + L_{\text{tv}} + L_{\text{exp}} + L_{\text{color}}$	**19.35**	**0.80**	**0.15**	**398.94**	**2.39**	**5.68**

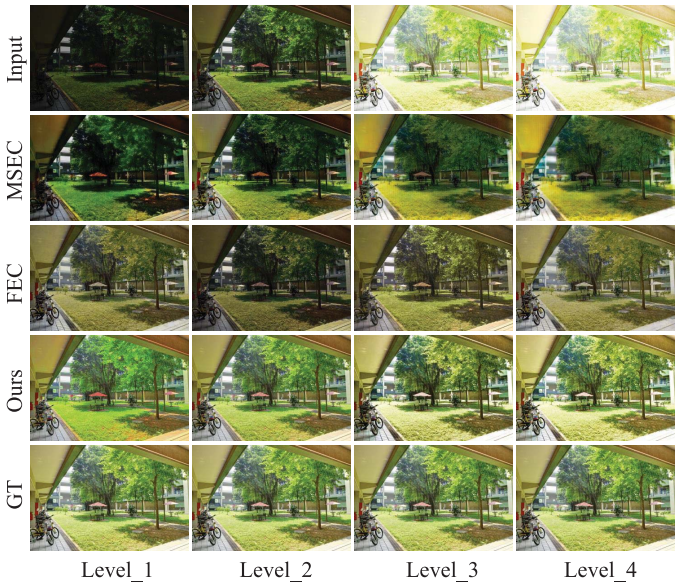

Fig. 12. Visual comparison in under/overexposed scenes.

2) ACT-Net setting: BAB, FDD, BH, and GAB: As can be seen in the fourth to seventh rows of Table VI, after using the BAB, the internal features of the adaptive adjustment decoder can perceive the brightness distribution and then more adaptively enhance the contrast of the image, which further improves the baseline performance by 0.330 on the PSNR. After equipping FDD, the performance of the baseline is further improved by 0.175 on PSNR. After equipping BH, the baseline network can perceive the image brightness distribution more efficiently, which further improves the baseline performance by 0.152 on PSNR. After equipping the GAB, the baseline can adaptively adjust the distribution of smoothness and detail features in different regions and improve the detailed edge of the low-light image, which further improves the baseline performance by 0.106 on the PSNR. We also perform the visualization results of the ablation studies on two low-light conditions, as shown in Fig. 9. After equipping the proposed framework and module sequentially, the contrast and details of the low-light images are gradually enhanced. It demonstrates the effectiveness of our proposed method.

To demonstrate the effectiveness of the proposed RF, we show the intermediate enhancement results in the recursive enhancement phase under two different low-light scenes, as shown in Fig. 11. One can observe that the recursive enhancement times of Level_1 image predicted by BP-Net are more than those of the Level_2 image. With the recursive enhancement of ACT-Net, the brightness distribution of the enhanced image is continuously improved until it reaches normal exposure. It shows that ACT-Net can recursively enhance image detail and contrast, and BP-Net can perceive the brightness distribution to guide ACT-Net to perform the adaptive enhancement.

3) Loss function: To explore the impact of loss functions used in the manuscript, we perform an ablation study on loss functions. Since the network cannot be trained without L_p, this section conducts ablation experiments for illumination smoothness loss L_{tv}, exposure control loss L_{exp}, and color constancy loss L_{color}. We provide more numerical results by applying different loss functions for training. As shown in Table VII, it is hard to effectively enhance the low-light image by only utilizing L_p and L_{tv} for training, which could cause all indicators to be significantly worse than using all losses. When utilizing L_p and L_{exp}, the PSNR and SSIM performances improve by 4.65 and 0.05. After applying all loss functions simultaneously, the model achieves the best performance.

F. Extending to Multiexposure Correction

In order to demonstrate the generalization performance and effectiveness of our proposed method, we compare our method with existing multiple-exposure methods (i.e., FEC [1] and MSEC [35]) in under and over-exposure scenes. Specifically,

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we calculate the average brightness of each image on the V channel in HSV color space and then divide these images according to the images' average brightness range from [0, 0.25), [0.25, 0.5), [0.5, 0.75) to [0.75, 1), denoted as Level 1, Level 2, Level 3, and Level 4, respectively. Then, we use this dataset to train FEC, MSEC, and our methods, and the results are shown in Fig. 12. It is easy to observe that for under-exposure Level 1 and Level 2 images, FEC and MSEC have the problem of under-enhancement, leading to the under-exposure of the enhanced image. For over-exposure Level 3 and Level 4 images, FEC and MSEC have the problem of color distortion, while our method can adaptively enhance both under-exposure and over-exposure images.

V. Conclusion

In this article, we proposed a novel unsupervised recursive enhancement framework to adaptively enhance image detail and contrast in low-light images with wide dynamic ranges. Specifically, a novel ACT-Net was proposed to enhance the details and brightness of low-light images simultaneously, which utilizes the brightness adjustment branch and gradient adjustment branch to balance image contrast and detail enhancement. To adaptively enhance low-light images, a simple but effective BP-Net was proposed to perceive the brightness distribution of low-light images and predict recursive factors to control the recursive enhancement times of ACT-Net, avoiding the under or over-exposure problems. Experiments demonstrated that the proposed method achieves SOTA performance in five image quality metrics.

In future work, we will consider adding a denoise module to remove the inevitable remaining noise in our enhanced result to further improve image quality. Besides, we will also utilize more real low-light images for training our unsupervised method to improve robustness in various low-light environments.

Data Availability Statement

The related code will be released at https://github.com/7407810/Brightness-Perceiving-for-Recursive-LLIE.

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